

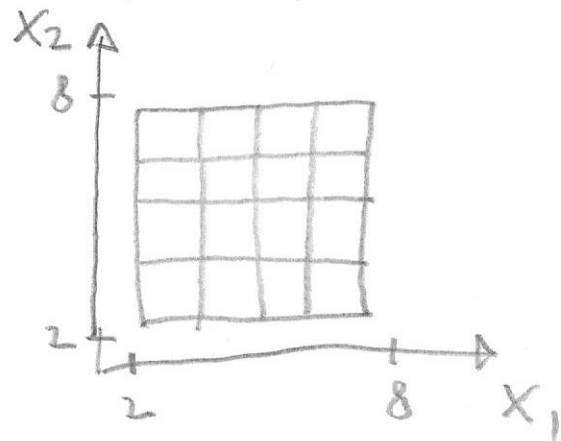
(1) Output with $\vec{w}_1 = (2, 3)$ wins.

$$\vec{w}_1 \cdot \vec{x} = 2 \cdot 1 + 3 \cdot 4 = 14$$

$$\vec{w}_2 \cdot \vec{x} = 3 \cdot 1 + 2 \cdot 4 = 11$$

(2) To maximise output $y = \vec{w} \cdot \vec{x}$

(3) input space



output space

7.9				7.9
2.1				7.9
		5.0		
		5.0		
2.1				2.1
2.1				7.9

etc.

(4) 2D $\rightarrow$ 2D - see (3) above

1D $\rightarrow$ 2D - bot

2D $\rightarrow$ 1D - travelling salesman